FINAL PROJECT PROPOSAL · DATA ENGINEERING × APPLIED AI

SkillNexus

Career intelligence for Vietnamese IT job seekers. Every answer comes from real market data, and every number shows its source.

Market truth → My gap → My plan → My CV

Team	[Name] Data Engineer · [Name] Data Scientist & Knowledge Graph · [Name] Applied AI & Full-stack
Mentor	[Name]
Duration	8 weeks · tentatively 26 Oct – 20 Dec 2026
Version	Draft 1 · 22 Oct 2026

01 Executive summary

Vietnamese IT job seekers decide what to learn, what to ask for and how to write their CV with little real evidence. Job posts are inflated and often hide the salary, and the market reports come out once a year as PDFs. AI CV tools rewrite text against generic standards, not against what Vietnamese employers actually ask for.

SkillNexus is a data platform plus an AI layer. A daily pipeline collects and normalizes Vietnamese IT job data, annual salary reports and open datasets into a warehouse where **every figure carries its source, period and sample size**. On top of that, local LLM workflows tell a user:

- which skills the market wants for their target role;
- what those skills are worth;
- how much of the demand their CV already covers;
- what to learn next, in what order, within their weekly time;
- how to tailor their CV to a specific job post **without inventing experience**.

The project is designed to teach a complete **data engineering** workflow (Airflow, a medallion warehouse, dbt, a graph projection), real **data science** (co-occurrence analysis, a job-post inflation index, salary-premium regression, trend momentum), and **applied AI** (LLM-in-ETL extraction, a guarded agent workflow, a tool-using market analyst with evaluation sets). It still addresses a problem people would pay to solve.

02 Problem statement

We profiled the 794 software and data job posts in **VietJobs**, an open dataset of 48,092 Vietnamese job ads collected from TopCV in July 2025 (LREC 2026). The results match the complaints developers repeat in community groups.

9.4%

of software/data posts accept candidates with **under 1 year** of experience

VietJobs dev subset, n = 794 · our analysis

74.8%

require **2+ years** of experience

Same subset · "Senior-ized Fresher" effect

56.9%

hide the salary ("Thỏa thuận" / 0)

Same subset; 28.5% across all sectors (paper)

4.4×

Back-end median salary from < 1 yr (12.4M VND) to > 8 yrs (54.9M VND)

ITviec IT Salary Report 2025-26, n = 1,839

Pain point	Who feels it	Evidence
P1 · "What does the market actually want?" Job posts list long, inflated stacks; nobody shows what is *commonly* required for a role, level and city.	Students, freshers	Only 9.4% of dev posts are open to < 1 yr experience; around 50,000 IT graduates a year, about 30% considered job-ready (Viet Nam News, 2025).
P2 · "What am I worth, and what would raise it?" Salaries are hidden or vague; reports are yearly PDFs by role, not by skill.	Juniors, mid-level switchers	56.9% of dev posts hide the salary; compensation transparency is the most cited reason to switch jobs (internal research digest, to validate).
P3 · "Why does nobody reply to my CV?" CVs list tutorial projects and keywords that don't match real posts; AI rewriters invent experience.	All levels	Mass CV sending ("rải CV") and ChatGPT keyword-stuffing are the common workarounds (developer community research digest).
P4 · "What should I learn next, in what order?" Static roadmaps ignore the local market and the user's time budget.	Freshers, career switchers	The shortage of 150k–200k developers a year since 2023 coexists with unemployed graduates: a skill-matching problem, not a headcount problem.

Market-level figures from secondary press coverage and internal AI-generated research digests are marked as such and will be re-verified before the defense.

03 Existing solutions & gap

Product	Market data for candidates	Skill gap vs. real VN posts	Personal roadmap	CV tailoring grounded in market data	Provenance on numbers
ITviec (job board + yearly salary report)	yearly PDF	—	—	—	report-level
TopCV (CV builder, Toppy AI scoring)	—	generic	—	generic	—
TopDev (job board + yearly market report)	yearly PDF	—	—	—	report-level
LinkedIn Premium (global, $39.99/mo)	global	global	—	global	—
roadmap.sh / CareerViet CareerMap	—	—	static	—	—
ChatGPT / generic AI CV tools	—	—	generic	no grounding	—
"AI Job Market Consultant" (academic prototype, arXiv 2511.14767: ReAct agent over 3,745 VN IT posts)	✓	partial	—	—	—
SkillNexus	✓ live, sliceable	✓	✓ time-budgeted	✓ + claim check	✓ every number

> **Where SkillNexus fits:** no Vietnamese product links *market evidence* → *personal gap* → *plan* → *CV* on one shared, traceable dataset. Our edge over the closest academic work is time-aware data with provenance, salary modelling per skill, and the end-to-end personal chain.

Competitor capabilities are based on public desk research (Oct 2026) and will be re-checked before the defense.

04 Proposed solution

Four modules form one user journey. Each module reads from the same gold tables, so the market numbers a user sees in module 1 are the same ones that rank their gaps in module 3.

M1 · Market Pulse

"What does the market want?"

- Skill demand by role × level × city
- Skill bundles (co-occurrence graph)
- **Job-post inflation index**
- Salary transparency index
- Ask-the-market agent with citations

M2 · Trends & Salary

"What is rising? What am I worth?"

- Digitized annual reports → momentum (rising / flat / falling)
- Salary range for a profile
- **Salary premium per skill** (with CI)
- Real vs nominal wages (CPI)

M3 · My Gap & Plan

"How far am I, and what next?"

- Market-coverage score of my CV
- Ranked missing skills
- Ordered roadmap within my hours per week
- Capstone project + free resources

M4 · My CV

"How do I present it for this post?"

- CV tailored to a specific post
- Batch tailoring for saved posts
- **Claim check**: nothing the CV can't back up
- Diff view with reasons

Design principles

1. **Math computes, the LLM narrates.** Every number comes from SQL, a graph query or a model; the LLM explains it and writes text.
2. **Provenance on every number:** source · period · sample size · confidence. Values with n < 30 are hidden by default.
3. **No fake trends.** Trend lines only connect points from the same source series; no numeric forecasts in the MVP.
4. **Honest scores.** "Covers 62% of skill demand across 412 Junior Backend posts", never "62% chance of being hired".
5. **Privacy first.** CVs are processed by a local LLM only and never mixed into shared market data.

05 Data strategy

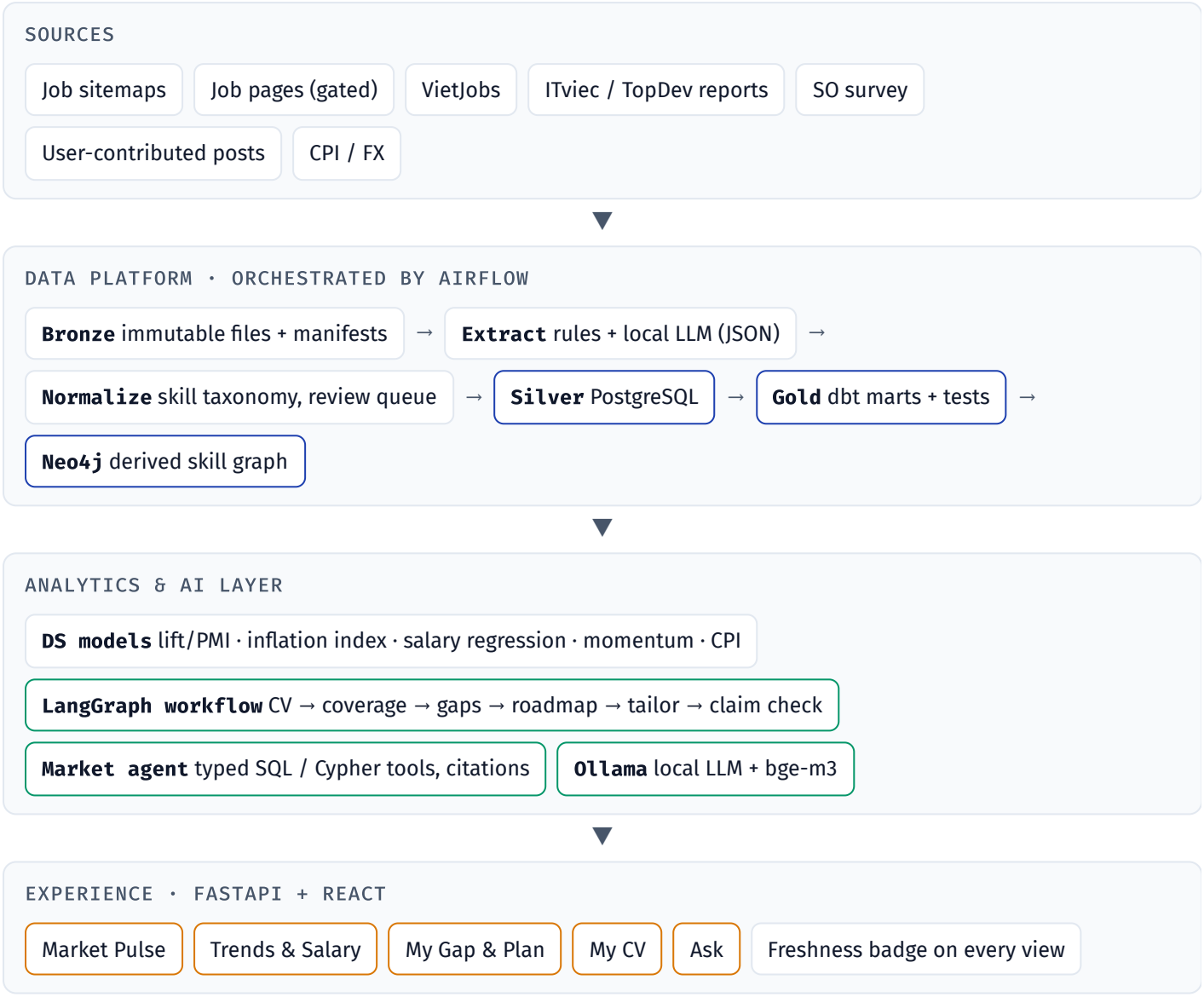
We start with backfill and published reports, then add a **light, rule-respecting live feed**: job sitemaps, which the sites publish for crawlers, plus fact extraction from new job pages once the team signs off on the sites' terms.

Source	What it gives	Time axis	Role in SkillNexus
Job sitemaps: ITviec (~640 live IT posts), TopDev (~2,780 mixed)	Job URLs, titles from URL slugs	Daily snapshots → first-seen / last-seen	Dated posting counts; feeds detail extraction `week 1`
Job detail pages (new posts only, rate-limited)	Skills, level, experience, salary, city (facts only)	Daily	Main live corpus, est. 1.5k–3k IT posts by week 8 `gated`
VietJobs (HF, LREC 2026)	~794 software/data posts; 342 with salary	One snapshot (Jul 2025)	Seed for the skill taxonomy, baseline point, salary model data

ITviec salary reports 2022-23 → 2025-26	Survey salaries by role × experience	4 yearly points	Salary truth + momentum
TopDev market reports 2021 → 2024-25	Market size, popular tech, demand	Yearly	Long-range context + momentum
Stack Overflow Developer Survey (ODbL)	Global tech usage and pay	Yearly since 2011	Global baseline
User-contributed posts (opt-in)	Posts users paste for tailoring	Live	Data flywheel without crawling
World Bank / GSO	CPI, USD↔VND	Monthly	Real-wage adjustment

Key research finding: VietJobs is often described as "48k IT jobs", but it contains only **1,906** IT-category rows and about **794** real software/data posts. That is enough for a seed set and a baseline, not for role × level × city statistics. So the light live feed is on the **critical path**, and the job-detail fact extraction depends on a terms-of-use sign-off (gate G1, end of week 1). If G1 is refused, we fall back to user-contributed posts plus an optional licensed API (jobspipe.dev, ≈ $20/month) and narrow the scope of market statistics.

06 System architecture



Technology stack

Layer	Choice	Why
Orchestration	Apache Airflow (LocalExecutor)	Standard DE orchestrator; retries, backfills, human-verification gates
Storage	Local bronze files · PostgreSQL 16 (silver/gold/app) · Neo4j 5 Community (derived)	SQL for time-window analytics; graph only for bundles and learning paths
Transformation & quality	dbt-core + dbt tests	Lineage, tests, documentation
Data science	pandas/polars, statsmodels, scikit-learn, Jupyter	Regression with confidence intervals; EDA as deliverables
LLM & embeddings	Ollama; model chosen by benchmark (Qwen2.5-7B-Instruct / Qwen3-8B / SEA-LION-8B); bge-m3	Local for privacy and zero API cost; multilingual for Vietnamese

AI orchestration	LangGraph: one deterministic workflow + one tool-using agent	Testable and reliable on 7–8B models
Parsing	Docling (CVs, report tables)	Layout-aware PDF tables
App	FastAPI · React + TypeScript + Tailwind	Already scaffolded in the repo

Full design: `architecture-overview.md` (data model, DAGs, agent guardrails, ADRs). Removed from the earlier draft: MinIO (community edition retired in 2025), Redis, Kafka/CDC, and four autonomous agents.

07 Scope

IN · MVP

- Daily pipeline: sitemaps → job facts → silver → gold → graph
- Backfill: VietJobs, 4 ITviec reports, TopDev reports, SO survey, CPI/FX
- M1: demand, bundles, inflation index, transparency, market agent
- M2: report momentum, salary range & skill premium, real wages
- M3: coverage score, ranked gaps, time-budgeted roadmap
- M4: JD-tailored CV with claim check (+ batch)
- Evaluation harness with 4 gold sets
- PDPL controls: consent, local processing, deletion

STRETCH

- Job-post decoder (must-have vs nice-to-have, red flags)
- Emerging-skill detector
- Hidden-salary imputation vs. the VietJobs paper baseline
- Weekly market newsletter; curriculum-gap report (B2B)

OUT

- Numeric forecasts (need ≥ 12 monthly points)
- Ghost-job detection (needs months of sightings)
- Company reviews / OT culture (legal risk)
- Auto-applying to jobs (terms-of-use and licensing risk)
- Mock-interview agent, payments, mobile app

08 Team roles & work division

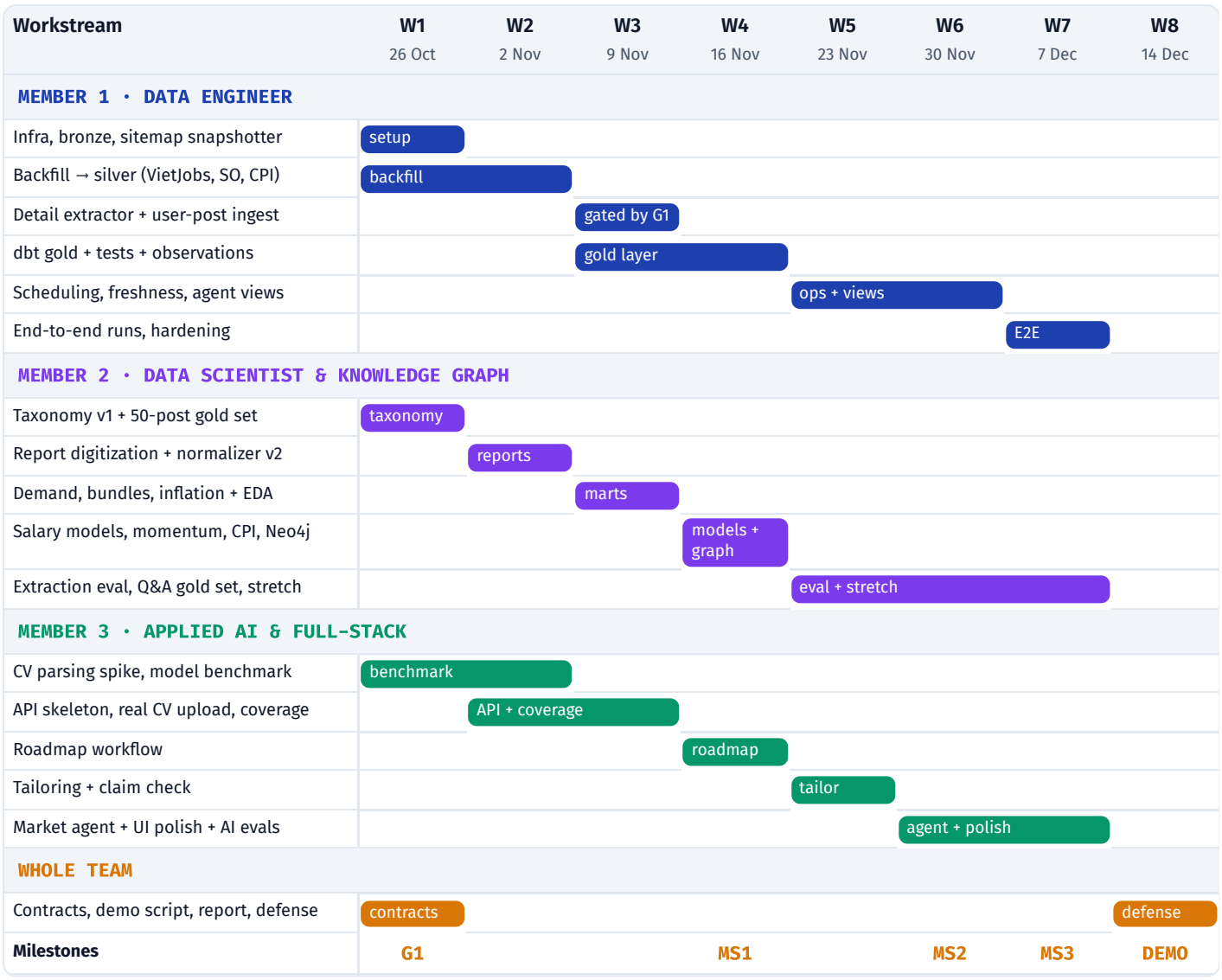
Member	Owns	Learns
Member 1 **Data Engineer** [Name]	Docker/infra and DB roles · Airflow DAGs for all sources · sitemap snapshotter and polite detail fetcher · bronze/silver layers · dbt gold marts and data tests · freshness API · read-only views for the agent	Orchestration, medallion modelling, idempotent ingestion, data quality, dbt
Member 2 **Data Scientist & Knowledge Graph** [Name]	Skill taxonomy and entity resolution · LLM extraction prompts and extraction evals · report digitization with human verification · analytics models (co-occurrence, inflation index, salary regression, momentum, CPI) · Neo4j projection and Cypher templates · EDA notebooks	Entity resolution, statistical modelling with uncertainty, graph analytics, LLM-in-ETL
Member 3 **Applied AI & Full-stack** [Name]	CV parsing · LangGraph personal workflow (coverage, gaps, roadmap, tailoring, claim check) · ask-the-market agent and guardrails · model benchmark · FastAPI endpoints · React UI · AI evals	Agent design, tool use, guardrails, evaluation, product UI

Responsibility matrix

Workstream	M1 DE	M2 DS/KG	M3 AI/App
Infrastructure & ingestion	**R**	C	I
Silver / gold / data tests	**R**	C	I
Taxonomy, extraction, report digitization	C	**R**	I
Analytics models & graph	C	**R**	C
CV workflow, roadmap, tailoring	I	C	**R**
Market agent	C	C	**R**
API & UI	I	I	**R**
Evaluation & gold sets	C	**R** data	**R** AI
Report, slides, demo	**R**	**R**	**R**

R = responsible · C = consulted · I = informed. Shared data contracts (Pydantic schemas in `contracts/`) are agreed in week 1, so all three members can build in parallel against mock data.

09 Timeline (8 weeks)



Milestone	End of	Exit criteria
G1 · Data audit & go/no-go	W1	Sitemap snapshots running daily; VietJobs profiled; report PDFs collected; **terms-of-use decision for detail extraction**; model shortlist
MS1 · Market data live	W4	Gold marts rebuilt daily from one command; demand, bundles and inflation shown in the UI with provenance; salary models and momentum computed
MS2 · Feature complete	W6	CV → coverage → roadmap → tailored CV works end to end; claim check passes on gold CVs
MS3 · Evaluated	W7	All four gold-set evaluations reported; market agent ≥ 80% accurate; pipeline success ≥ 95% over the last 7 days
DEMO · Defense	W8	Demo script rehearsed, report and slides done, ~6 weeks of our own dated posting data

Dates assume a start on Monday 26 October 2026 and shift with the approval date. Week 8 is deliberately kept as buffer.

10 Evaluation & success criteria

Area	Metric	Target
Data volume	Dated IT postings with extracted skills	≥ 1,500 (if G1 approved) · otherwise ≥ 800 + reports
Pipeline reliability	Daily DAG success rate (last 7 days) · dbt tests	≥ 95% · 100% passing
Normalization	Skill mentions resolved automatically	≥ 90%
Extraction quality	Skill F1 on a 50-post hand-labelled gold set · field accuracy	≥ 0.85 · ≥ 0.90
CV parsing	Skill F1 on 10 anonymized CVs	≥ 0.85
Trustworthiness	Fabricated claims in tailored CVs (10 CVs × 2 posts)	**0**
Market agent	Accuracy on 30 questions with SQL-computed answers · answers with citations	≥ 80% · 100%
Salary model	Out-of-sample RMSE / R^2, compared with the VietJobs paper's published LLM baselines (best R^2 = 0.16)	Reported transparently; aim to beat the baseline
Experience	CV upload → roadmap, on a 16 GB laptop	≤ 60 s, streamed

11 Risks & mitigations

Risk	Likelihood / impact	Mitigation
Terms-of-use sign-off for job detail pages is refused	Medium / High	Sitemap-only statistics; user-contributed posts; optional licensed API; narrower market scope (decided at G1)
Too little posting data for fine segments	Medium / Medium	n ≥ 30 rule, coarser segments (role × level), confidence labels instead of hidden gaps
Local 7–8B model too weak or slow for Vietnamese extraction	Medium / High	Rules first (VietJobs skills are already structured); benchmark in W2; overnight batches; optional cloud batch for *public* data only
Annual reports change categories between editions	High / Medium	Per-edition mapping table; compare only within one publisher; human-verified cells
LLM rewrites invent experience	Medium / High	Claim check, diff view, "to learn" list; zero-fabrication target tested on gold CVs
One laptop runs out of RAM	Medium / Medium	MinIO/Redis removed; Neo4j heap reduced; heavy jobs scheduled apart (~10–12 GB total)
Scope creep across 59 brainstormed features	High / Medium	Frozen MVP list; documented minimum cut of 8 features; week 8 kept as buffer

12 Legal, privacy & ethics

- **Personal data (PDPL, Law 91/2025/QH15, in force since 1 Jan 2026):** explicit, unbundled consent per upload; CVs processed only by a local LLM, so there's no cross-border transfer; raw files deleted within 24 h; one-click full deletion; user data never enters the shared market dataset.
- **Third-party job content:** the sites' `robots.txt` files allow job pages, and we only fetch what they permit, at a low rate with an honest user-agent. We store **extracted facts, not copied text**, and purge raw pages after 30 days. Use is non-commercial and academic; a commercial launch would require partnerships or a licensed feed. TopDev's terms forbid copying and redistributing content; ITviec's terms are still to be reviewed (gate G1).
- **Reports and open data:** every figure taken from ITviec or TopDev reports is cited per edition. Stack Overflow data follows ODbL attribution. The VietJobs dataset license is to be confirmed.
- **Employment-service licensing:** SkillNexus does not place candidates or forward applications, so it stays an analytics tool, not a licensed recruitment intermediary.
- **Fairness & honesty:** no hiring-probability claims; confidence labels; recommendations never encourage misrepresentation.

13 Budget & business value

Project budget

Item	Cost
Infrastructure (local Docker on team laptops)	0 VND
LLM inference (Ollama, local)	0 VND
Data: open datasets, reports, sitemaps	0 VND
Optional: jobspipe.dev, 1 month	≈ $20
Optional: cloud batch LLM for public data	< $5
Required total	**0 VND** (≤ ~$25 optional)

Business model (post-MVP)

- **Free:** Market Pulse and limited market questions, which also drive acquisition and community sharing.
- **Pro (~79,000 VND/month):** roadmap, unlimited and batch CV tailoring, salary-premium detail. Priced inside the $3–8/month range Vietnamese developers report being willing to pay (to validate).
- **B2B:** curriculum-vs-market gap reports for universities and training centers, built from the same gold tables.
- **Growth:** an auto-generated weekly market newsletter for developer communities.

Learning outcomes

Field	What the team will be able to show
Data engineering	Scheduled, idempotent multi-source ingestion; medallion warehouse; dbt models and tests; data contracts; freshness monitoring
Data science	Entity resolution; co-occurrence lift/PMI; a data-driven inflation index; salary regression with confidence intervals; honest trend handling across mixed granularities
Applied AI	LLM-in-ETL with validation; agent workflow with guardrails; tool-using agent with citations; offline evaluation with gold sets; local-LLM model selection

14 Decisions & inputs needed

#	Decision / input	Needed by	From
1	Approve the terms-of-use approach for job detail extraction (gate G1)	End of W1	Team + mentor
2	Download report PDFs that sit behind sign-up forms (ITviec / TopDev)	W1	Any member
3	Hardware per member (RAM / GPU), which fixes the local model choice	W1	All members
4	Confirm the VietJobs dataset license	W1	Member 2

5	Volunteers for 10 anonymized CVs (consent forms)	W2	Member 3
6	Optional: jobspipe.dev sign-up to test Vietnam coverage	W2	Member 1
7	Official start date, team names, grading-rubric weights	Before W1	Team lead + mentor

15 References

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